

Report: Fine-Tuning vs. From-Scratch Training for Text Classification

1. Introduction

This report compares two approaches for training decoder-only language models on the **AG News** text classification task:

- 1. **From-Scratch Training:** A small, custom Transformer decoder (~3.4M parameters).
- 2. **Fine-Tuning:** A pre-trained `roneneldan/TinyStories-33M` model (~68.5M parameters).

The objective is to analyze the trade-offs between performance (accuracy/F1) and efficiency (training/inference speed, model size).

2. Dataset Description

Dataset: `AG News` **Task:** Topic Classification (4 classes: World, Sports, Business, Sci/Tech).

2.1 Statistics & Split

The dataset was split into Train, Validation, and Test sets. A 10% validation split was created from the original training set.

Split	Number of Examples
Train	108,000
Validation	12,000
Test	7,600

2.2 Class Distribution

The dataset is perfectly balanced, ensuring that accuracy is a reliable metric.

- **Train:** ~27,000 examples per class.
- **Validation:** ~3,000 examples per class.

2.3 Preprocessing

- **Tokenization:** Both models utilized the `roneneldan/TinyStories-33M` tokenizer (GPT-2 based BPE) to ensure a fair comparison of vocabulary.
- **Sequence Length:** Truncated to 128 tokens to manage memory constraints.
- **Class Mapping:**
 - 0: World
 - 1: Sports
 - 2: Business
 - 3: Sci/Tech

3. Model Descriptions

3.1 From-Scratch Model (Small Decoder)

A custom-built decoder-only Transformer designed to be lightweight.

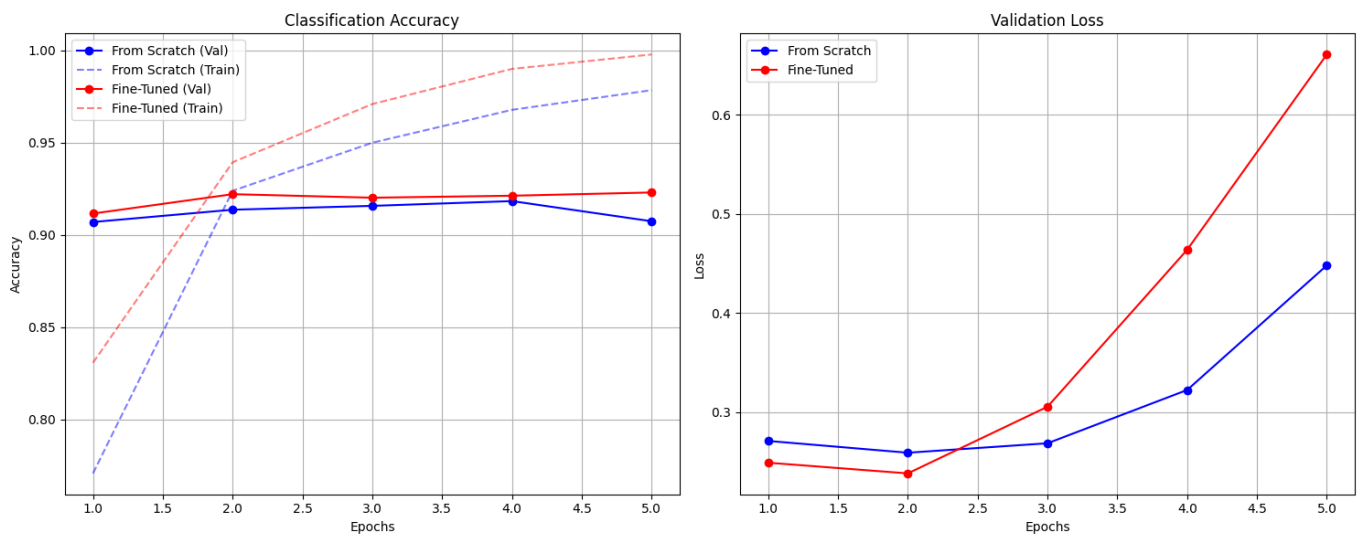
- **Architecture:** `TransformerClassifier` (Custom `nn.Module`)
- **Type:** Decoder-only (Causal masking applied).
- **Classification Head:** Mean pooling over the sequence followed by a 2-layer MLP.
- **Configuration:**
 - Embedding Dimension: 64
 - Hidden Size: 256
 - Heads: 4
 - Layers: 4
 - Dropout: 0.1
- **Parameter Count: 3,434,052 (~3.4M)**

3.2 Fine-Tuned Model (Pre-trained)

- **Base Model:** `roneneldan/TinyStories-33M` (Hugging Face).
- **Architecture:** GPT-Neo style decoder-only transformer.
- **Pre-training:** Trained on a synthetic dataset of short stories containing only words that a 3-4 year old typically understands.
- **Classification Head:** Linear layer on top of the last token's hidden state.
- **Parameter Count: 68,517,120 (~68.5M)**

4. Training & Evaluation Results

4.1 Training Dynamics



Both models were trained using the AdamW optimizer with a linear warmup. Early stopping was set with a patience of 3 epochs.

- **From-Scratch:** Trained for 5 epochs. Best validation loss was achieved at **Epoch 2** (0.2590).
- **Fine-Tuned:** Trained for 5 epochs. Best validation loss was achieved at **Epoch 2** (0.2381).

Observations from Curves:

- **Convergence:** The fine-tuned model converged rapidly, reaching peak performance almost immediately (Epoch 2).
- **Overfitting:** Both models exhibited signs of overfitting after Epoch 2, where validation loss began to increase while training loss continued to decrease. The fine-tuned model's validation loss degraded more sharply (from 0.23 to 0.66), suggesting catastrophic forgetting or simple overfitting to the training noise given its larger capacity.

4.2 Quantitative Results Table

Metric	From-Scratch (Small)	Fine-Tuned (TinyStories)
Test Accuracy	91.51%	92.18%
Test F1 Score	91.44%	92.16%
Parameters	~3.4M	~68.5M
Total Train Time	~11.6 min (697s)	~90.4 min (5422s)
Inference Time	2.3s	22.8s
Speedup Factor	10x faster	1x

5. Comparative Analysis

5.1 Performance vs. Size

The most striking result is how competitive the **From-Scratch** model is. Despite having **20x fewer parameters** (3.4M vs 68.5M), it achieved an accuracy of **91.51%**, which is less than 1% lower than the Fine-Tuned model (**92.18%**).

Why was the From-Scratch model so effective?

1. **Task Simplicity:** AG News is a topic classification task with distinct vocabularies (e.g., "touchdown" for Sports vs "stocks" for Business). It does not require complex reasoning or world knowledge, which is where pre-trained models typically excel.
2. **Sufficient Capacity:** 3.4M parameters proved sufficient to learn the statistical patterns of this specific dataset.

5.2 Efficiency & Speed

The From-Scratch model dominates in efficiency:

- **Training:** It took ~11 minutes to train, compared to ~1.5 hours for the fine-tuned model on the same hardware.
- **Inference:** It is **10x faster** at inference (2.3s vs 22.8s for the test set). This makes the small scratch model significantly more viable for real-time applications or deployment on edge devices (mobile phones, IoT).

5.3 Stability & Overfitting

- **Fine-Tuned Instability:** The pre-trained model overfit aggressively after just 2 epochs. The validation loss spiked from 0.23 to 0.66. This indicates that even a "small" LLM like TinyStories (68M) has enough capacity to memorize the training data quickly, requiring aggressive regularization (higher dropout, weight decay) or fewer epochs.
- **Scratch Stability:** The scratch model also overfit, but the degradation was more gradual.

6. Conclusion

This experiment demonstrates that **bigger is not always better**, especially for specific, narrower tasks like topic classification.

- **Fine-Tuning** provided the absolute best metrics, leveraging its pre-trained knowledge of language structure.
- **From-Scratch** training provided a highly efficient, lightweight alternative that sacrificed negligible accuracy (<0.7%) for a **10x gain in inference speed** and a massive reduction in memory footprint.